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<tr><td>MIT | Academy of Engineering
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Alandi (D), Pune – 412105</td><td colspan="2">SYNOPSIS</td></tr>
<tr><td>DEPARTMENT OF ELCTRONICS AND TELECOMMUNICATION ENGINEERING</td><td>ACADEMIC YEAR</td><td>2025-2026</td></tr>
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1. **Project Title :- IoT-based Smart Agriculture System for Soil Condition Monitoring and Crop Prediction**

2. **Members:-**

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3. **Introduction :-**

   Agriculture is the foundation of food security and rural livelihood. One of the key challenges farmers face is selecting the right crop for their land based on soil and environmental conditions. Wrong crop choices often result in reduced yield, soil degradation, and financial losses.

   With the advent of the Internet of Things (IoT) and Artificial Intelligence (AI), it has become possible to monitor soil and environmental parameters in real time and provide data-driven recommendations.

   This project, **Smart Agriculture System**, focuses on monitoring critical parameters such as **soil moisture, pH level, and temperature** using IoT-enabled sensors. The collected data is processed and analyzed to **predict the most suitable crops** for the given environment. The system also integrates cloud connectivity, enabling farmers to view real-time soil data and crop recommendations on dashboards or mobile devices.

   This solution aims to empower farmers with scientific decision-making tools, enhance productivity, and promote sustainable agricultural practices.

4. **Objective(s):-**

   - To design and implement an IoT-based system for monitoring soil and environmental parameters (moisture, pH, and temperature).

   - To develop a decision-making model that predicts suitable crops based on collected data.

- To integrate cloud platforms for real-time monitoring, storage, and visualization.
- To promote sustainable agriculture by guiding farmers toward optimal crop selection.

5. SDG Goal(s) Addressed:-

1. SDG 2: Zero Hunger – Improves crop yield through intelligent recommendations and disease detection, ensuring food security.
2. SDG 3: Good Health and Well-being – Promotes healthier food production and reduces harmful farming practices.
3. SDG 6: Clean Water and Sanitation – Enables smart irrigation to minimize water wastage and ensure sustainable water use.
4. SDG 9: Industry, Innovation, and Infrastructure – Introduces IoT, AI, and cloud technologies to modernize agriculture infrastructure.
5. SDG 12: Responsible Consumption and Production – Encourages efficient use of natural resources and sustainable farming.SDG 13: Climate Action – Supports climate-resilient agriculture and reduces environmental impact.

6. Project Description / Methodology:-

The Smart Agriculture System follows a structured IoT-based approach combining sensing, decision-making, and cloud connectivity.

Step 1: Sensor Data Collection

- Soil Moisture Sensor measures soil water content.
- DHT22 sensor records temperature and humidity.
- Light sensor (LDR/BH1750) monitors sunlight intensity.
- Data is continuously read by the ESP8266 microcontroller.

Step 2: Data Preprocessing

- Raw sensor values are filtered and calibrated to avoid false readings.
- Soil moisture , pH and temperature ranges are standardized
- Threshold values (e.g., minimum soil moisture level) are predefined based on crop requirements.

Step 3: Crop Prediction Model

- If Collected data is compared with crop requirement datasets.
- A rule-based or ML-based algorithm predicts the most suitable crops for the current environment.

Step 4: Cloud Connectivity

- The Wi-Fi (ESP8266) transmits data to cloud platforms (ThingSpeak, Firebase, or MQTT).
- Dashboards allow farmers to monitor soil conditions and receive crop suggestions.

Step 5: Alerts and Recommendations

- Wi-Fi Farmers receive real-time recommendations via a mobile/PC interface.
- Suggestions include top 2–3 crops suited to current soil and climate conditions.

Step 6: Feedback and Optimization

- Historical data is analyzed to improve accuracy of predictions.
- Future enhancements include integrating AI/ML models for advanced crop forecasting.

7. Expected Outcomes:-

- **Data-Driven Crop Selection:** Farmers receive accurate crop recommendations.
- **Improved Productivity:** Matching crops with soil conditions enhances yield.
- **Sustainability:** Prevents soil degradation by avoiding unsuitable crops.
- **Technology Adoption in Farming:** Encourages IoT and AI integration in agriculture.
- **Scalable Model:** System can be applied to different crops, soil types, and regions.

8. Impact and Sustainability:-

The Smart Agriculture System has the potential to create a significant long-term impact by transforming traditional farming into a technology-driven, efficient, and sustainable practice.

Long-Term Impact:-

- Increased Agricultural Productivity: Automated irrigation ensures that crops receive optimal water supply, improving yield and food security.
- Reduced Water Wastage: Smart water management directly addresses water scarcity challenges, preserving freshwater resources for future generations.
- Economic Benefits for Farmers: Lower water and labor costs improve profit margins, especially for small-scale farmers.
- Improved Rural Livelihoods: Farmers gain access to affordable, modern agricultural solutions, bridging the gap between rural communities and advanced technologies.
- Scalable Model: The system can be deployed in different crop types and farm sizes, making it adaptable globally.

Sustainability:-

- Environmental Sustainability: By reducing over-irrigation, the project conserves water, reduces soil erosion, and minimizes environmental damage.
- Energy Efficiency: The system can be powered using solar panels, making it eco-friendly and suitable for remote rural areas.
- Technological Sustainability: Cloud integration allows continuous data logging and analysis, enabling AI/ML upgrades for predictive farming.
- Alignment with SDGs: The project supports SDG 2 (Zero Hunger), SDG 6 (Clean Water), SDG 12 (Responsible Consumption), and SDG 13 (Climate Action), ensuring long-term contributions to sustainable development.

9. Resources Required :-

Hardware Components:-

- ESP8266 – Microcontroller for sensor interfacing and control
- Soil Moisture Sensor – To measure soil water content

- DHT22 Sensor – For temperature and humidity monitoring

- pH Sensor – To measure soil acidity/alkalinity.
- ESP8266 Wi-Fi Module / LoRa Module – For cloud connectivity
- Power Supply (Battery / Solar Panel) – To power the system in the field

Software & Platforms:-

- Arduino ide
- Firebase– For cloud storage and visualization
- Mobile App – For dashboards and remote monitoring

Other Materials:-

- Breadboard, jumper wires, resistors, and basic electronic components
- Enclosure box to protect hardware from outdoor conditions

Support Required:-

- Access to agricultural field for testing and deployment
- Internet connectivity for real-time cloud monitoring
- Guidance from faculty/mentors for system integration and testing

10. Timeline:-

Phase	Activities	Duration
Phase 1: Research & Planning	Literature review, identifying problems in traditional irrigation, selecting SDGs, and finalizing project objectives.	Week 1
Phase 2: Component Selection & Procurement	Choosing hardware (ESP8266, sensors, relay, Wi-Fi module) and required software tools.	Week 2
Phase 3: System Design	Circuit design, flowchart preparation, and algorithm/decision logic development.	Week 3
Phase 4: Implementation	Hardware connections, Arduino coding, and integrating sensors with microcontroller.	Weeks 4–5
Phase 5: Cloud & Networking Setup	Configuring ESP8266/LoRa, setting up ThingSpeak/Blynk dashboard, and enabling data transmission.	Week 6
Phase 6: Testing & Debugging	Testing sensor readings, cloud updates, and fixing issues.	Weeks 7–8
Phase 7: Data Analysis & Optimization	Analyzing collected data, refining threshold values, and ensuring reliable automation.	Week 9
Phase 8: Documentation & Presentation	Preparing project report, poster, and final presentation.	Week 10

11. Conclusion:-

The proposed Smart Agriculture System enables **real-time monitoring of soil moisture, pH, and temperature** and provides **intelligent crop recommendations** to farmers. By leveraging IoT and AI technologies, the system helps improve agricultural productivity, reduces risks of unsuitable crop selection, and supports sustainable farming practices. This project demonstrates how digital solutions can empower farmers, ensure food security, and contribute to global sustainable development goals.